



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/34

Paper 3 Further Mechanics

October/November 2025

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2}

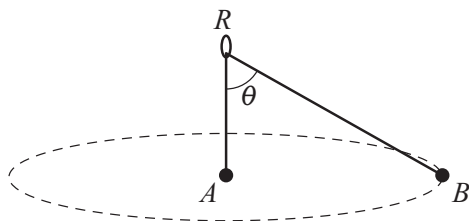
INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.



1



A light inextensible string of length $12a$ is threaded through a fixed smooth ring R . One end of the string is attached to a particle A of mass m . The other end of the string is attached to a particle B of mass $0.5m$. Particle A hangs in equilibrium vertically below the ring. Particle B moves with constant angular speed ω in a horizontal circle with particle A at its centre. The angle between AR and BR is θ (see diagram).

Express ω in terms of g and a .

[5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple rows of horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings.



- 2 A particle P is projected with speed $u \text{ m s}^{-1}$ at an angle θ , where $\tan \theta = 2$, above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at a time $t \text{ s}$ are denoted by $x \text{ m}$ and $y \text{ m}$ respectively.

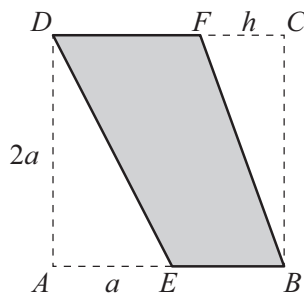
(a) Use the equation of the trajectory given in the list of formulae (MF 19) to show that

$$y = 2x - \frac{25x^2}{u^2} . \quad [1]$$

- (b) In the subsequent motion, P passes through the point with coordinates $(8, 12)$. The particle then hits a fixed vertical barrier 7 m high that is at a horizontal distance of D m from the point of projection.

Find the set of possible values of D . [5]





The lamina $BFDE$ is obtained by removing triangles AED and BCF from a uniform square lamina $ABCD$ of side $2a$. The length of side AE is a and the length of side FC is h (see diagram). The centre of mass of $BFDE$ is at a distance $\bar{x}$ from AD , and at a distance $\bar{y}$ from AB .

- (a) Show that $\bar{x} = \frac{h^2 - 6ah + 11a^2}{3(3a - h)}$ and find a corresponding expression for $\bar{y}$. [5]

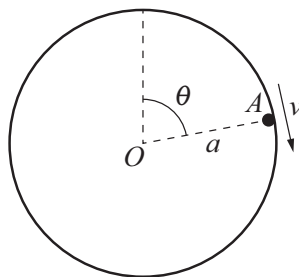
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- (b)** The lamina $BFDE$ is placed vertically on its edge EB on a smooth horizontal surface.

Find, in terms of a , the set of possible values of h for which the lamina remains in equilibrium. [2]

[illegible]



A fixed smooth spherical shell has centre O and radius a . A particle of mass m moves in complete vertical circles on the smooth inner surface of the shell, where the plane of the circular motion is vertical and passes through O . The particle has speed v when it is at point A , where OA makes an angle θ with the upward vertical through O , and $\cos \theta = \frac{1}{18}$ (see diagram).

- (a) Show that $v \geq \frac{1}{3}\sqrt{26ag}$. [5]

This image shows a full page of a document template designed for handwritten notes or essays. It features approximately 28 evenly spaced, thin grey horizontal lines across the entire width of the page. The margins are consistent on all sides, providing a clear area for writing. There are no other markings, text, or graphics present on the page.



It is given that $v = \frac{1}{3}\sqrt{26ag}$.

- (b) Find, in terms of m and g , an expression for the greatest possible value of the normal reaction between the shell and the particle. [3]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.



- 5 A particle P of mass m kg is projected vertically upwards from a point O with an initial speed of 20 m s^{-1} and moves under gravity. There is a resistive force of magnitude $0.025mv^2 \text{ N}$, where $v \text{ m s}^{-1}$ is the speed of P at time t s after projection. The displacement of P from O is x m at time t s after projection.

(a) Find an expression for v in terms of x , while P is moving upwards. [5]

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(b) Find an expression for v in terms of t , while P is moving upwards.

[4]

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- 6 A and B are two fixed points at a distance $22a$ apart, with B vertically below A . A light elastic string of natural length $4a$ and modulus of elasticity $5mg$ has one end attached to A and the other end attached to a particle P of mass km . Another light elastic string of natural length $8a$ and modulus of elasticity $6mg$ has one end attached to B and the other end attached to P . Particle P is vertically above B .

(a) Show that, when the system is in equilibrium, $BP = \frac{57a - 2ak}{4}$. [5]

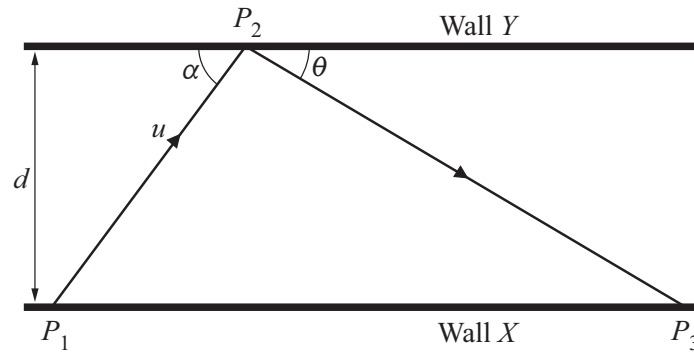
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The particle P is pulled vertically upwards so that $BP = 18a$, and is then released from rest. In its subsequent motion, P first comes to instantaneous rest at the point where $BP = 8a$.

(b) Find the value of k . [4]

[illegible]



X and Y are two fixed smooth vertical walls on a smooth horizontal surface. The walls are parallel and at a distance d apart. The points P_1 , P_2 and P_3 all lie on the surface.

A particle Q is projected horizontally from the point P_1 on Wall X with speed u , and moves along the surface. The particle Q strikes Wall Y at the point P_2 . Immediately before the collision, the direction of motion of Q makes an angle α with Wall Y , where $\sin \alpha = \frac{4}{5}$. Immediately after the collision, the direction of motion of Q makes an angle θ with Wall Y . The particle Q then strikes Wall X at the point P_3 (see diagram).

The time that it takes Q to travel the distance P_1P_2 is T . The time that it takes Q to travel the distance P_2P_3 is kT .

Find, in terms of k , the coefficient of restitution between Q and Wall Y . [6]

[illegible]

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Additional page

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